

PMS

Assignment - 3

Arpit Kumar

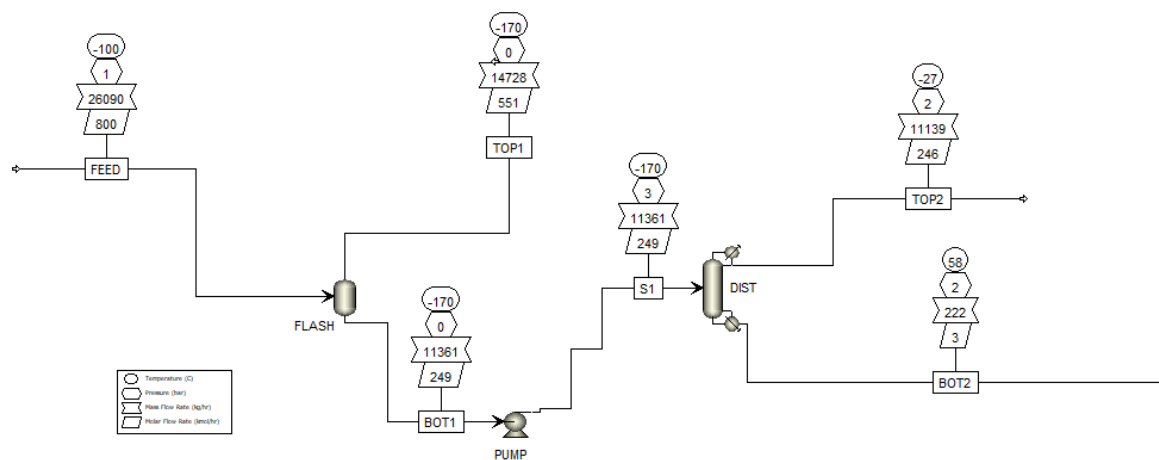
16th March, 2025

Answer 1:

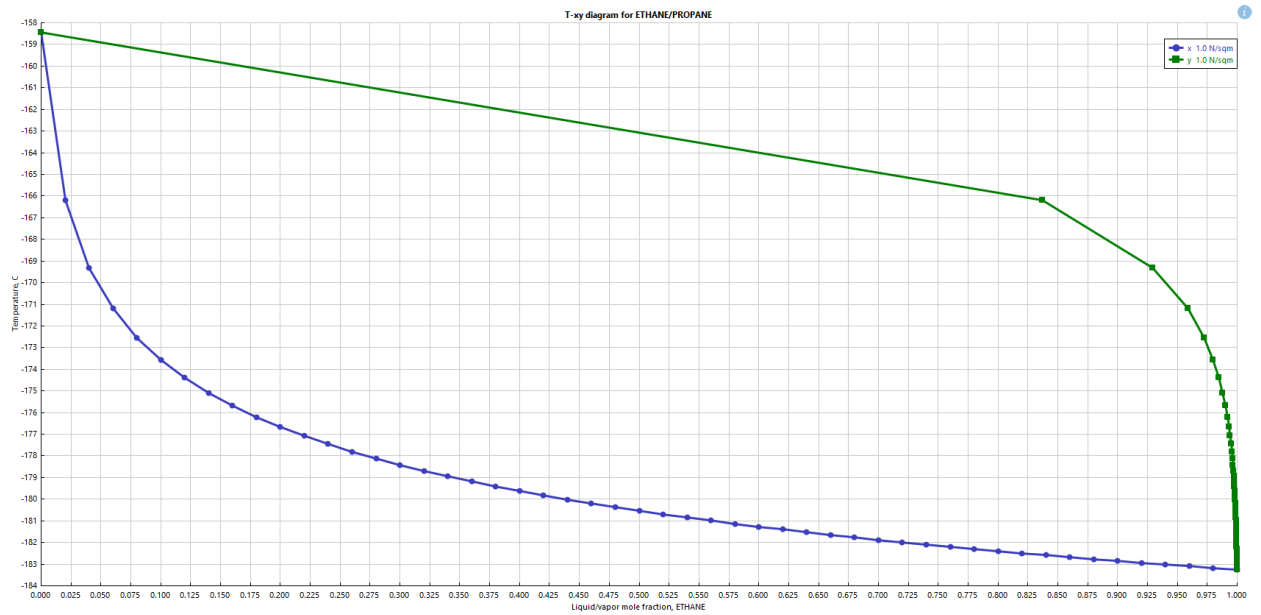
- **Purpose:** To determine the sequencing of distillation columns for the given feed composition.
- **Objective:** To establish the type of condenser and operating pressure for two different sequences:
 - **Direct sequence (C2/C3 separation first)**
 - **Indirect sequence (C3/nC4 separation first)**

For separation of the C2/C3 (direct separation)

- **Flowsheet for Direct Sequence (C2/C3 separation first)**



- **Binary Analysis of C2/C3**

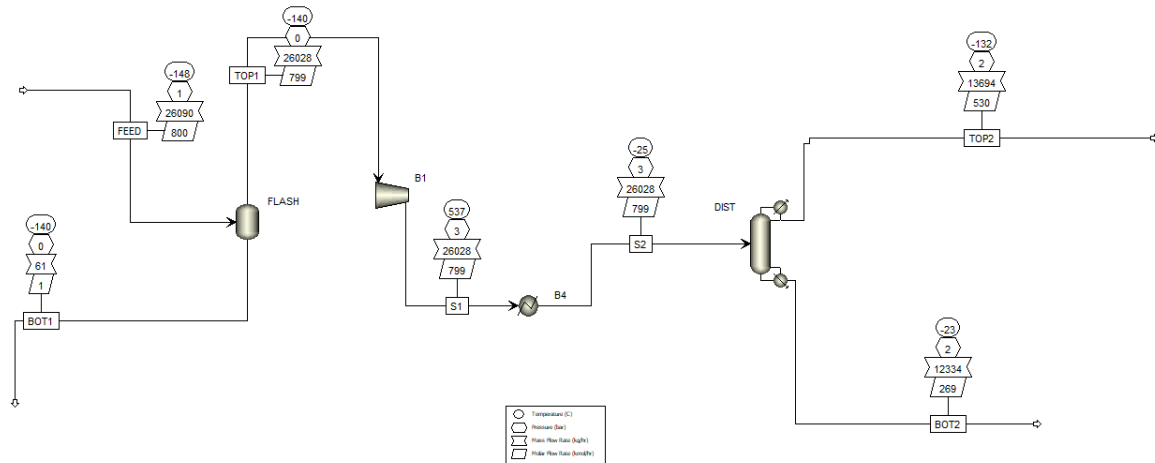


- By Binary analysis plot of C2/C3 we get a range of flash temperature to operate

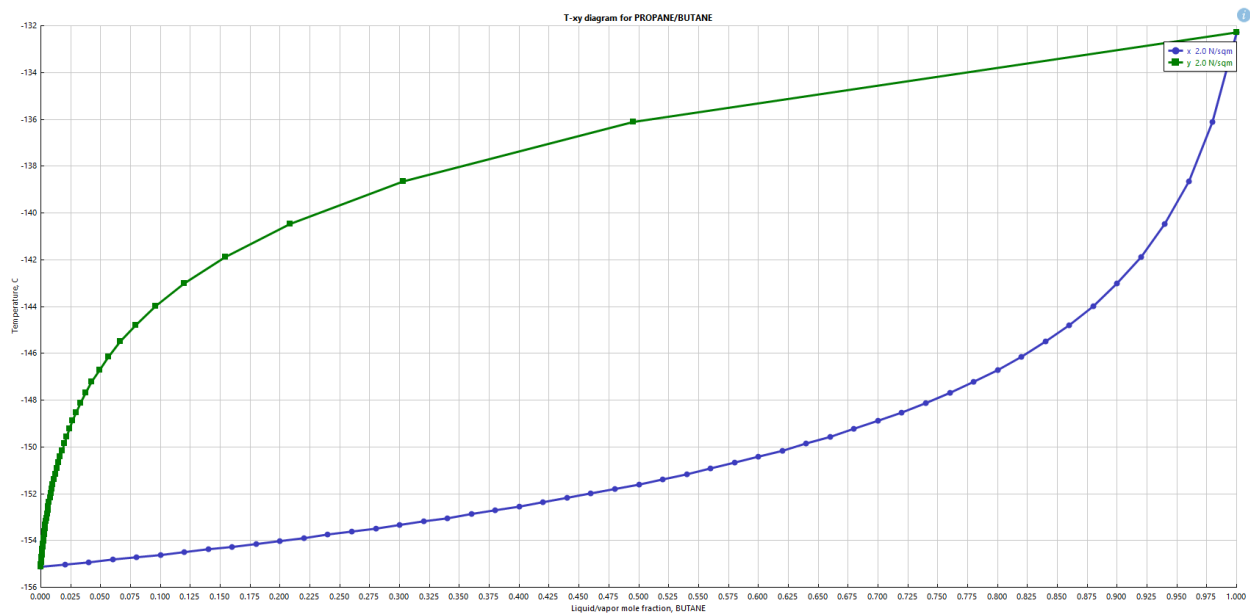
Temperature range for flash = **-165 to -175 °C**

For separation of the C3/C4 (Indirect separation)

- Flowsheet for Indirect Sequence (C3/C4 separation first)



- Binary Analysis of C3/C4

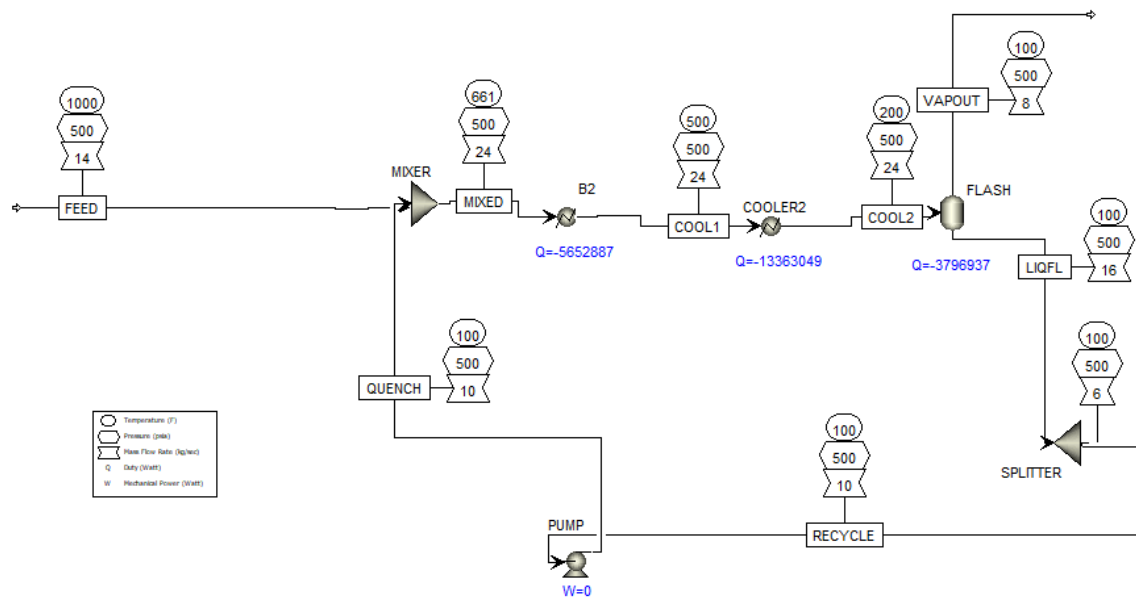


- By Binary analysis plot of C3/C4 we get a range of flash temperature to operate

Temperature range for flash = **-142 to -146 °C**

Answer 2:

- Flowsheet for system to cool reactor effluent and separate light gases from hydrocarbons



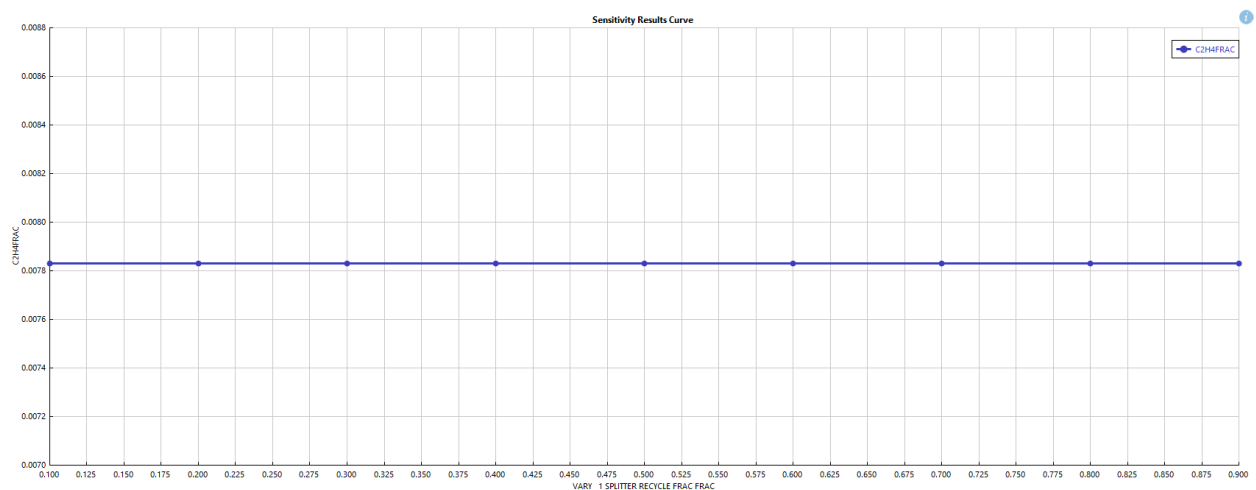
- Calculated composition and flow rate of vapor leaving the flash drum.

Mole Flows	lbmol/hr	3859.922024
HYDROGEN	lbmol/hr	1994.035061
ETHANE	lbmol/hr	1833.087501
BENZENE	lbmol/hr	30.22104805
TOLUENE	lbmol/hr	2.578413786
Mole Fractions		
HYDROGEN		0.51659983
ETHANE		0.474902729

BENZENE		0.007829445
TOLUENE		0.000667996
Mass Flows	kg/sec	7.778891057
HYDROGEN	kg/sec	0.506478141
ETHANE	kg/sec	6.945038614
BENZENE	kg/sec	0.297440181
TOLUENE	kg/sec	0.029934121
Mass Fractions		
HYDROGEN		0.065109299
ETHANE		0.892805744
BENZENE		0.038236836
TOLUENE		0.003848122
Volume Flow	cum/sec	0.350629291

Impact of Quench on Flash Separation

- Figure of quenching analysis



- The **quench reduces temperature** before flashing but does not affect **phase equilibrium at 500 psia**.
- The **vapor-to-liquid ratio remains nearly the same**, confirming that quenching aids cooling but **does not influence phase separation**.
- The **molar flow rates and mole fractions** of the vapor phase remain unchanged.
- The quench stream improves **cooling efficiency** but has **minimal impact on flash separation results**.

Answer 5 :

Q.5. (I)

The Rate of production of H₂ is 0.112 kg/sec and The rate of production of O₂ is 0.895 kg/sec.

Here the Water that is going to Anode and Cathode is about 333.75 kg/sec, in which the In anode the water inlet is 250.313 kg/sec and Cathode inlet is 83.4375 kg/sec , so at the end the outlet of water at anode is 251.321 kg/sec and Cathode outlet for water is 81.4205 kg/sec.

From the data:

FEED (Total Inlet): 333.75 kg/sec (H₂O)

ANOUT (Outlet): 251.321 kg/sec (H₂O)

CAOUT (Outlet): 81.4205 kg/sec (H₂O)

The total water leaving the system is the sum of the water in the ANOUT and CAOUT streams:

Total Outlet Water = ANOUT (H₂O) + CAOUT (H₂O) = 251.321 kg/sec + 81.4205 kg/sec = 332.7415 kg/sec

The rate of water consumption is the difference between the total inlet water and the total outlet water:

Water Consumption = FEED (H₂O) - Total Outlet Water = 333.75 kg/sec - 332.7415 kg/sec = 1.0085

kg/sec

So, the rate of water consumption is approximately 1.0085 kg/sec.

From the table:

(Inlet): 250.313 kg/sec

(Outlet): 251.321 kg/sec

The makeup water required would be the difference between the outlet and inlet water flow rates:

Makeup Water = ANOUT (H₂O) - ANIN (H₂O) = 251.321 kg/sec - 250.313 kg/sec = 1.008 kg/sec

So, the makeup water required is approximately 1.008 kg/sec.

ii

Rigorous Method

Reaction Components

Henry Components

a. Electrolyzer Efficiency

The efficiency of an electrolyzer can be calculated using the formula:

Efficiency = [(Energy content of hydrogen produced) / (Electrical energy input)] × 100%

The energy content of hydrogen is typically given by its lower heating value (LHV), which is approximately 33.33 kWh/kg.

From the provided data:

Total power input = 20 MW

Number of stacks = 8

Number of cells per stack = 230

First, calculate the total electrical energy input per second:

$$\text{Electrical energy input} = 20 \text{ MW} = 20 \times 10^6 \text{ W} = 20 \times 10^6 \text{ J/s}$$

Next, determine the mass flow rate of hydrogen produced. From the table, the mass flow rate of H_2

at the cathode is 0.116429 kg/s.

Now, calculate the energy content of the hydrogen produced per second:

$$\text{Energy content of hydrogen} = 0.116429 \text{ kg/s} \times 33.33 \text{ kWh/kg} \times 3600 \text{ s/h}$$

$$\text{Energy content of hydrogen} \approx 13999.8 \text{ kJ/s}$$

Now, calculate the efficiency:

$$\text{Efficiency} = \left[\frac{13999.8}{20 \times 10^3} \right] \times 100\% \approx 70\%$$

b. Energy Requirements per Unit Mass of H_2 Production

The energy requirement per unit mass of hydrogen production can be calculated as:

$$\text{Energy requirement} = \text{Electrical energy input} / \text{Mass flow rate of hydrogen}$$

Using the values from above:

$$\text{Energy requirement} = 20 \times 10^6 / 0.116429 \approx 171.8 \text{ MJ/kg}$$

This value can also be expressed in kWh/kg:

$$\text{Energy requirement} = 171.8 / 3.6 \approx 47.7 \text{ kWh/kg}$$

(III) Flowsheet as follows:-

The molar mass of H_2 is approximately 2 g/mol or 0.002 kg/mol.

$$\text{Molar flow rate of } \text{H}_2 = 0.117006 / 0.002 = 58.503 \text{ mol/s}$$

Since the purity of H_2 is 96%, the actual molar flow rate of H_2 produced is:



Actual Mass flow rate of $H_2 = 58.503/0.96 = 60.94 \text{ mol/s}$

According to the stoichiometry of the reaction, 1 mole of water is required to produce 1 mole of H_2 .

So, Molar flow rate of $H_2O = 60.94 \text{ mol/s}$

The molar mass of water (H_2O) is approximately 18 g/mol or 0.018 kg/mol.

Mass Flow rate of $H_2O = 60.94 \times 0.018 = 1.096 \text{ kg/s}$